

European Power Fair Value Prototype Report

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Configuration Summary:

Market: [DE] Germany

Approach: [Option A] next-day hourly Day-Ahead prices

OOS Window: 1st October 2025 - 29th August 2026

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Section 1: Data Ingestion, QA, & Feature Engineering

1.1 Data Ingestion & Sourcing

Source	Raw Frequency	Processes Frequency	Core Extracted Features	Economic/Fundamental Justification	QA Implemented
ENTSO-E	1hr	Hourly	Target Price (Price_EUR_MWh)	TARGET	- Prevents pipeline crashes during 503 Service Unavailable throttling events. - Writes automated verification text files to verify dataset compactness - Identifies and logs extreme power grid events to flag anomalies.
SMARD	1hr	Hourly	Total_MWh: 122, PV_and_Wind_MWh: 5097, Wind_Offshore_MWh: 3791, Wind_Onshore_MWh: 123, Photovoltaics_MWh: 125, Other_MWh: 124, Forecasted_Grid_Load_(Forecasted_Total_Consumption)_MWh : 411	Direct representation of the physical supply/demand stack.	- Detects missing hourly data fragments caused by server timeout drops. - Flags and clips corrupted negative energy telemetry values. - Prevents time-series alignment skewing and downsampling leaks.
Open-Meteo	1hr	Hourly	FOR ONSHORE AND OFFSHORE: "temperature_2m", "wind_speed_10m", "wind_direction_10m", "wind_gusts_10m"	Acts as a proxy for upcoming renewable wind power penetration.	- Eliminates catastrophic linear averaging errors for Wind Direction. - Eliminates mismatch boundaries across individual 30-day API call loops - Protects onshore temperature/wind speeds from marine variance distortion.
yFinance	Daily	Hourly (Forward-filled)	Closing prices of: Natural gas (Dutch): TTF=F, Coal Prices: MTF=F, EU-EAS carbon : "CO2.L	Sets the marginal price floor for fossil-fuel generation (merit-order curve).	- Preserves essential weekend grid rows when global financial exchanges close. - Strict Data Leakage Protection (Look-Ahead Bias elimination). - Eliminates remaining empty rows from asynchronous holiday closures.

1.2 Strategic Feature Engineering

5 most important Features Engineered:

- **forecast_residual_load_mwh** : Power prices do not scale linearly with total load; they scale with *residual load* (Total Load minus intermittent Solar & Wind). This captures where the market moves up or down the thermal dispatch merit order curve. **HENCE A CORE RESIDUAL METRICE**
 - **gas_generation_cost_floor_eur** :It anchors the price to a physical baseline. By taking shifted Gas and Carbon data and applying standard efficiency metrics (50% efficiency for Gas units, 0.202 carbon intensity), it creates a rolling marginal cost floor without data leakage. **HENCE A STRUCTURAL ECONOMIC MARGINAL COST FLOOR**
 - **price_lag_24h & price_lag_168h**: Day-ahead power markets exhibit massive daily and weekly seasonality. Yesterday's price at hour and last week's price at hour act as the **BASELINE ANCHORS FOR THE MODEL**
 - **solar_saturation_index & regime_saturation_flag**: When solar generation crosses a critical threshold (approx. at 70% of grid load), the market frequently risks entering zero or negative pricing regimes. Explicitly mapping this non-linear threshold helps tree-based models (like LightGBM) split the data accurately.
 - **demand_ramp_3h_mw & wind_speed_volatility_delta**: Power prices spike during aggressive changes in demand or sudden drop-offs in wind. The 3-hour demand ramp and the 24-hour shift in wind variance capture the physical strain on system flexibility.
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Section 2: Forecasting Methodology & Validation

2.1 Validation Strategy & Data Splits

To evaluate the models without look-ahead bias, a **strict chronological out-of-time validation framework** was implemented. Traditional randomized k-fold cross-validation is inappropriate for power markets due to high temporal dependency and autocorrelation. The dataset is segmented sequentially into three distinct windows:

- **Training Window**: 2024-01-01 to 2025-06-30 (Used to train base estimators and capture structural seasonal variations).
- **Validation Window**: 2025-07-01 to 2025-09-30 (Used for hyperparameter tuning and optimizing ensemble weights)
- **Out-of-Sample Test Window**: 2025-10-01 onwards (Held completely blind to evaluate ultimate generalization performance and generate the predictions.csv submission file).

2.2 Model Architecture & Selection Justification

Four structural variations were engineered and benchmarked to find the optimal trade-off between bias, variance, and non-linear adaptability:

1. **Linear Ridge Regression (Baseline)**: Chosen as a robust linear baseline. Power prices are heavily anchored by thermal generation fuel costs and physical load. A regularized linear model efficiently handles high-dimensional calendar dummies and penalizes multicollinearity among related drivers (e.g., Gas, Coal, and Carbon).
2. **Time-Series LightGBM (All Features)**: LightGBM is selected because day-ahead power markets exhibit massive non-linearities and threshold effects such as solar saturation leading to near-zero pricing regimes. It processes these complex splits at high computational speeds.
3. **LightGBM with LASSO Selection**: Engineered to test feature parsimony. A LASSO {L1 regularization} pipeline first filters out non-contributing features before passing the optimal subset to LightGBM, attempting to combat overfitting.

4. **Hybrid LGBM + CatBoost Ensemble (FINAL):** The champion model. It leverages the strengths of two distinct gradient-boosting frameworks. LightGBM excels at fast split optimization across numeric feature interactions, while CatBoost utilizes symmetric trees to reduce variance and handle remaining categorical structural features natively.

2.3 Experimental Performance & Analysis

The performance metrics across the distinct phases are summarized in the table below:

Model Name	Validation RMSE	Validation MAE	Test RMSE	Test MAE
Linear Ridge Regression	24.75	15.48	29.31	19.14
Time-Series LightGBM (All Features)	22.61	14.34	32.1	21.38
LightGBM with LASSO Selection	23.82	14.79	32.92	21.43
Hybrid LGBM + CatBoost Ensemble (FINAL)	21.65	13.33	27.14	16.8

2.4 Performance Diagnostics

- **The Overfitting Trap Exposed:** While the standard standalone LightGBM model performed exceptionally well on the validation set (RMSE: 22.61), it degraded sharply on the unseen out-of-sample test set (RMSE: 32.10). This indicates that the single-tree architecture overfitted to specific summer/early-autumn dynamics present in the validation window.
- **The Power of Regularization:** Interestingly, the Linear Ridge Regression model generalized better out-of-sample (Test RMSE: 29.31) than the standalone LightGBM. This validates that structural, cost-based economic features act as reliable anchors when market regimes shift.
- **Ensemble Dominance:** The **Hybrid LGBM + CatBoost Ensemble** successfully counteracted individual model decay. By blending the diverse tree architectures, it achieved the lowest error metrics across both validation and testing windows, squeezing the **Test MAE down to a highly competitive 16.80 EUR/MWh**. This architecture was locked in to produce the final predictions.csv deliverables.

Section 3: Prompt Curve Trading Translation

3.1 Day-Ahead-to-Curve Translation Methodology

The translation from an hourly profile to a tradable curve position relies on an **Arbitrage Premium Rule**:

- **Go Long Prompt Curve:** If $\Delta\text{Arbitrage} > +\text{Execution Threshold}$ (Forecast indicates the market is underpricing upcoming fundamentals).
- **Go Short Prompt Curve:** If $\Delta\text{Arbitrage} < -\text{Execution Threshold}$ (Forecast indicates the market is overpricing upcoming fundamentals).

3.2 Operational Guidance & Invalidation Triggers

Because the prompt curve prices the risk premium of future delivery, a model's statistical view can be structurally invalidated by real-time shocks. The position must be tightly monitored against these three specific triggers:

- **Severe Weather Forecast Shift (Wind/Solar Deltas):** If weather models abruptly shift within 48 hours of delivery, changing wind speeds by ≥ 5 km/h or solar irradiance across major German hubs, the renewable generation forecast fails, invalidating the residual load features.
 - **Unplanned Nuclear or Thermal Outages:** Sudden, unpriced structural supply-side failures (e.g., cooling water restrictions, interconnector trips, or unplanned French nuclear outages) will sharply shift the marginal clearing price upward, forcing an immediate short-position cover.
 - **Extreme Marginal Fuel Volatility:** If European Natural Gas (TTF) or Carbon (EUA) spot markets move beyond a 2σ historical rolling window during the week, the `gas_generation_cost_floor_eur` feature shifts mechanically, disrupting the model's static baseline anchor.
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Section 4: Programmatic AI/LLM Integration

4.1 Architecture & Implementation Design

To eliminate manual analysis overhead, an AI forecasting evaluator is embedded directly into the core production data pipeline .

This component runs inside the row-by-row data streaming loop immediately after the machine learning model exports its predictions.

For each designated evaluation interval, the script extracts real-time physical grid telemetry and machine learning price predictions to construct a highly specific contextual prompt. This payload is passed via a secure API client to the gpt-4o-mini model instance.

4.2 Strategic Operational Justification

Using a programmatic LLM wrapper within the data loop provides clear advantages:

- **Contextual Data Interpretation:** Tree-based architectures output numbers but cannot explain the physical causes behind grid strain. The LLM translates numbers into explicit risk assessments by linking weather telemetry with pricing direction.
- **Operational Automation:** This automation removes mechanical friction. Instead of an analyst manually compiling morning risk notes, the pipeline generates actionable insights alongside the numeric predictions.
- **Auditability & Compliance:** By saving raw prompt payloads and completed answers directly to `ai_logs/ai_pointer_collection_with_insight.csv`, the system maintains a complete audit trail for end-of-day risk compliance reviews.